PRACTICE SHEET 3

Conductors & Charge Distribution • Gauss's Law & Electric Flux • Electric Field & Potential due to Sheets/Plates

Time Allowed: 60 Mins **Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

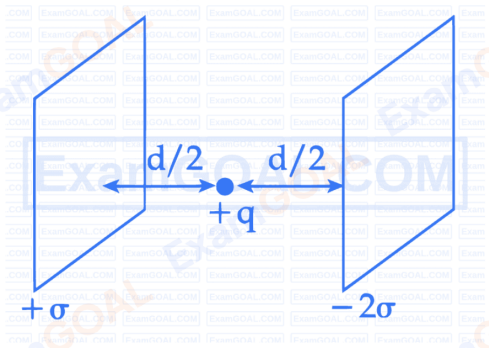
Instructions: Attempt all questions. Each question carries one correct option unless marked *one or more than one option is correct*. Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 Two metal plates (A, B) are kept horizontally with separation of $\left(\frac{12}{\pi}\right)$ cm, with plate A on the top. An atomizer jet sprays oil (density 1.5 g/cm^3) droplets of radius 1 mm horizontally. All oil droplets carry a charge 5 nC. The potentials V_A and V_B are required on plates A and B respectively in order to ensure the droplets do not descend. The values of V_A and V_B are _____. (Neglect the air resistance to the droplets and take $g = 10$ m/s^2) *[JEE Main 2026, 2 Apr, Eve.]*

(A) 100 V and 580 V (B) 580 V and 100 V

(C) 60 V and 400 V (D) 0 V and −200 V

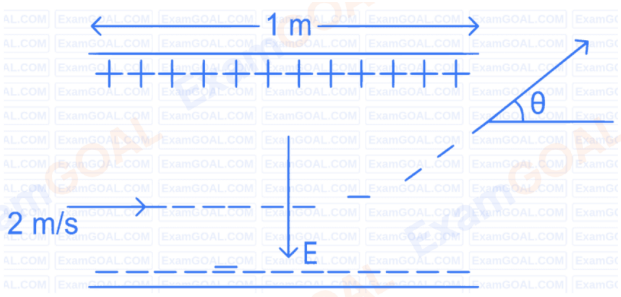
Q.2 Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ. The force experienced by a point charge $+q$ placed at the mid point between two plates will be: *[JEE Main 2025, 2 Apr, Mor.]*



(A) $\frac{3\sigma q}{4\epsilon_0}$ (B) $\frac{3\sigma q}{2\epsilon_0}$

(C) $\frac{\sigma q}{4\epsilon_0}$ (D) $\frac{\sigma q}{2\epsilon_0}$

Q.3 A uniform electric field $E = \left(8\frac{m}{e}\right)$ V/m is created between two parallel plates of length 1 m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of 2 m/s. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____. *[JEE Main 2022, 28 Jul, Eve.]*



(A) $\tan^{-1}(4)$ (B) $\tan^{-1}(2)$

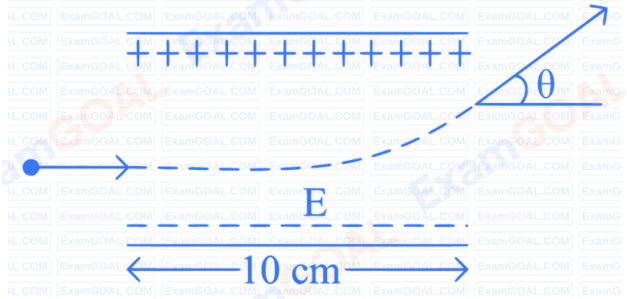
(C) $\tan^{-1}\left(\frac{1}{3}\right)$ (D) $\tan^{-1}(3)$

Q.4 A uniform electric field $\vec{E}$ exists between the plates of a charged condenser. A charged particle enters the space between the plates and perpendicular to $\vec{E}$. The path of the particle between the plates is a: *[JEE Main 2013, 9 Apr, Mor.]*

(A) straight line (B) hyperbola

(C) parabola (D) circle

Q.5 A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree). *[JEE Main 2023, 25 Jan, Mor.]*



Q.6 Two metal spheres of radius R and $3R$ have same surface charge density σ. If they are brought in contact and then separated, the surface charge density on smaller and bigger sphere becomes σ_1 and σ_2, respectively. The ratio $\frac{\sigma_1}{\sigma_2}$ is *[JEE Main 2025, 8 Apr, Eve.]*

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) 9 (D) 3

Q.7 A small uncharged conducting sphere is placed in contact with an identical sphere but having 4×10^{-8} C charge and then removed to a distance such that the force of repulsion between them is 9×10^{-3} N. The distance between them is (Take $\frac{1}{4\pi\epsilon_0}$ as 9×10^9 in SI units) *[JEE Main 2025, 24 Jan, Eve.]*

(A) 1 cm (B) 2 cm (C) 4 cm (D) 3 cm

Q.8 Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is: *[JEE Main 2024, 6 Apr, Eve.]*

(A) 1 N (B) 6 N (C) 12 N (D) 4 N

Q.9 The vehicles carrying inflammable fluids usually have metallic chains touching the ground: *[JEE Main 2024, 5 Apr, Eve.]*

(A) To protect tyres from catching dirt from ground
(B) It is a custom
(C) To alert other vehicles
(D) To conduct excess charge due to air friction to ground and prevent sparking

Q.10 Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is: *[JEE Main 2023, 30 Jan, Mor.]*

(A) $\frac{5}{3}$ (B) $\frac{5}{6}$ (C) $\frac{9}{4}$ (D) $\frac{4}{3}$

Q.11 Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have $12\,\mu$C and $-3\,\mu$C charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively: *[JEE Main 2020, 3 Sep, Mor.]*

(A) $4.5\,\mu$C on both (B) $+4.5\,\mu$C and $-4.5\,\mu$C
(C) $6\,\mu$C and $3\,\mu$C (D) $3\,\mu$C and $6\,\mu$C

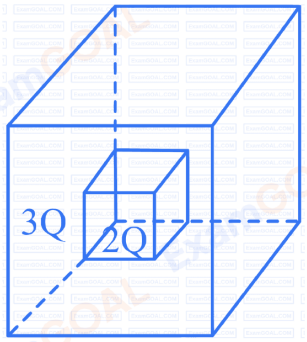
Q.12 Consider a finite insulated, uncharged conductor placed near a finite positively charged conductor. The uncharged body must have a potential: *[JEE Main 2013, 23 Apr, Mor.]*

(A) less than the charged conductor and more than at infinity.
(B) more than the charged conductor and less than at infinity.
(C) more than the charged conductor and more than at infinity.
(D) less than the charged conductor and less than at infinity.

Q.13 A point charge causes an electric flux of -2×10^4 Nm^2C^{-1} to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is: (Given $\epsilon_0 = 8.85 \times 10^{-12}$ $\text{C}^2\text{N}^{-1}\text{m}^{-2}$) *[JEE Main 2025, 29 Jan, Eve.]*

(A) 17.7×10^{-8} C (B) -17.7×10^{-8} C
(C) 15.7×10^{-8} C (D) -15.7×10^{-8} C

Q.14 C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is: *[JEE Main 2024, 1 Feb, Eve.]*



(A) 3 : 2 (B) 5 : 2 (C) 2 : 5 (D) 2 : 3

Q.15 An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k})$ N/C. The electric flux through a surface area $30\,\hat{i}$ m^2 lying in YZ-plane (in SI unit) is: *[JEE Main 2024, 29 Jan, Eve.]*

(A) 60 (B) 90 (C) 180 (D) 150

Q.16 In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the center of the surface S having area of $4L^2$. The flux through the opposite surface to S is given by *[JEE Main 2023, 29 Jan, Mor.]*

(A) $\frac{q}{2\epsilon_0}$ (B) $\frac{q}{3\epsilon_0}$
(C) $\frac{q}{12\epsilon_0}$ (D) $\frac{q}{6\epsilon_0}$

Q.17 Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius $4a$ having center at origin is: *[JEE Main 2024, 29 Jan, Mor.]*

(A) $\frac{2Q}{\epsilon_0}$ (B) $\frac{7Q}{\epsilon_0}$
(C) $\frac{3Q}{\epsilon_0}$ (D) $\frac{5Q}{\epsilon_0}$

Q.18 Choose the incorrect statement:

(1) The electric lines of force entering into a Gaussian surface provide negative flux.

(2) A charge q is placed at the centre of a cube. The flux through all the faces will be the same.

(3) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.

(4) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

Choose the most appropriate answer from the options given below: *[JEE Main 2021, 31 Aug, Eve.]*

(A) (3) and (4) only (B) (2) and (4) only
(C) (4) only (D) (1) and (3) only

Q.19 In finding the electric field using Gauss Law the formula $|\vec{E}| = \frac{q_{enc}}{\epsilon_0 |A|}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. The equation can be used in which of the following situation? *[JEE Main 2020, 8 Jan, Mor.]*

(A) Only when $|\vec{E}|$ = constant on the surface.
(B) For any choice of Gaussian surface.
(C) Only when the Gaussian surface is an equipotential surface.
(D) Only when the Gaussian surface is an equipotential surface and $|\vec{E}|$ is constant on the surface.

Q.20 If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2, the electric charge inside the surface will be *[AIEEE 2003]*

(A) $(\phi_2 - \phi_1)\epsilon_0$ (B) $(\phi_2 + \phi_1)/\epsilon_0$
(C) $(\phi_1 - \phi_2)/\epsilon_0$ (D) $(\phi_1 + \phi_2)\epsilon_0$

SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Electrostatics – Conductors, Gauss's Law & Plates

SELF-ASSESSMENT – PRACTICE SHEET 3

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____